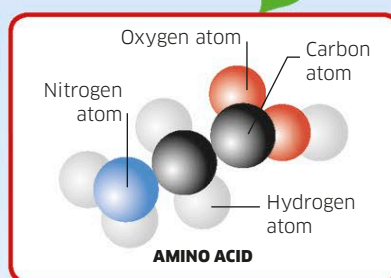


Cycles of matter

Many of Earth's crucial materials for life are constantly recycled through the environment.

All the atoms that make up the world around us are recycled in one way or another. Chemical reactions in living things, such as photosynthesis and respiration, drive much of this recycling. These processes help pass important elements like carbon and nitrogen between living things, the soil, and the atmosphere.

Nitrogen gas makes up about two-thirds of Earth's atmosphere.



Nitrogen in molecules

Molecules containing nitrogen—such as this amino acid—are used by plants, animals, and bacteria. It helps with growth and other vital functions.

Plants use nitrate from the roots to make food. When a leaf falls, it still contains this nitrogen.

The nitrogen cycle

Nitrogen exists in many forms inside living things, including in DNA, proteins, and amino acids. Animals and many bacteria obtain their nitrogen by feeding on other organisms—dead or alive. Plants absorb it as a mineral called nitrate—a chemical that gets released into the soil through the action of the bacteria.

Some kinds of bacteria turn nitrates into nitrogen gas, which is released into the atmosphere: a process called denitrification.

Lightning strikes can cause nitrogen gas to react with oxygen. This can release mineral nitrogen back into the soil—a process called nitrogen fixation.

Some kinds of bacteria help release minerals, such as nitrate, into the soil after feeding on dead leaves. This is called nitrification.

The dead and decaying matter of living things contains nitrogen.

Plants get their nitrogen by absorbing nitrate through their roots.

NITRATE

NITROGEN MOLECULES

Nitrogen-containing amino acids are in fallen leaves.

